Truncation level of the induction map along an n-connected map into a family of k-types

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:conn-trunc-variable-ind' (source lines 1569-1573); the proof environment begins on the next line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own section heading 'The Freudenthal suspension theorem' and the paragraph that motivates varying the truncation level, so the lemma appears under its original heading.

Editorial note. The retained counter values (chapter 8, section offset 5) let the retained section increment the counter to 6 and reset the theorem counter, so the counted lemma prints as Lemma 8.6.1, matching the original numbering.

Editorial note. One cited label contains a space in the author's own identifier (prop:nconnected_tested_by_lv_n_dependent types); it is reproduced verbatim both in the retained proof and in the source-reference record.

Editorial scope. One original result of the HoTT Book, counted once: Truncation level of the induction map along an n-connected map into a family of k-types.

Allowed standard prerequisites (editorial).

- n-connected maps and the fixed-level induction principle. A map f : A -> B is n-connected when all its fibres are n-connected; the book's Lemma 7.5.7 states that for fixed n, precomposition (blank o f) is an equivalence for every family P : B -> n-Type.. That lemma is the base case k = n of the retained induction (source line 1576) and is cited again for the family Q at source lines 1585-1588.
- Fibres, sigma types and paths between dependent pairs. The fibre of a map g at a point is Sm{x}(g(x) = point); elements of a sigma type are equal as soon as their components are, with the second component compared after transport.. The proof computes the fibre of the precomposition map at source line 1580 and the identity type of two of its elements at source lines 1581-1584.
- Truncatedness of path spaces and of function types into truncated families. If P is a family of (k+1)-types then b |-> (g(b) = h(b)) is a family of k-types; and a type is m-truncated as soon as all its path spaces are (m-1)-truncated.. These two facts close the induction step at source lines 1588-1589.

Source reference. Original section 7: sources/hott-homotopy/book-labelnumbers-first-edition.tex, source lines 166–166. Pointer to the chapter where the fixed-level facts about n-connected maps are proved.

Source reference. Original lemma 7.5.7: sources/hott-homotopy/book-labelnumbers-first-edition.tex, source lines 1570–1570. Pointer to the fixed-level induction principle for n-connected maps, which is the base case of the retained induction; the induction step that constitutes the content of the counted lemma is retained in full at source lines 1575-1589.

8.6 The Freudenthal suspension theorem

Before proving the Freudenthal suspension theorem, we need some auxiliary lemmas about connectedness. In original section 7 we proved a number of facts about n-connected maps and n-types for fixed n; here we are now interested in what happens when we vary n. For instance, in original lemma 7.5.7 we showed that n-connected maps are characterized by an "induction principle" relative to families of n-types. If we want to "induct along" an n-connected map into a family of k-types for $k > n$, we don't immediately know that there is a function by such an induction principle, but the following lemma says that at least our ignorance can be quantified.

Lemma 8.6.1. *If $f : A \to B$ is n-connected and $P : B \to k\text{-}\mathsf{Type}$ is a family of k-types for $k \geq n$, then the induced function*

$$(- \circ f) : \Big(\prod_{b:B} P(b)\Big) \to \Big(\prod_{a:A} P(f(a))\Big)$$

is $(k - n - 2)$-truncated.

Proof. We induct on the natural number $k - n$. When $k = n$, this is original lemma 7.5.7. For the inductive step, suppose f is n-connected and P is a family of $(k + 1)$-types. To show that $(- \circ f)$ is $(k - n - 1)$-truncated, let $\ell : \prod_{(a:A)} P(f(a))$; then we have

$$\mathsf{fib}_{(- \circ f)}(\ell) \simeq \sum_{(g:\prod_{(b:B)} P(b))} \prod_{(a:A)} g(f(a)) = \ell(a).$$

Let (g, p) and (h, q) lie in this type, so $p : g \circ f \sim \ell$ and $q : h \circ f \sim \ell$; then we also have

$$\big((g, p) = (h, q)\big) \simeq \Big(\sum_{r:g\sim h} r \circ f = p \cdot q^{-1}\Big).$$

However, here the right-hand side is a fiber of the map

$$(- \circ f) : \Big(\prod_{b:B} Q(b)\Big) \to \Big(\prod_{a:A} Q(f(a))\Big)$$

where $Q(b) :\equiv (g(b) = h(b))$. Since P is a family of $(k+1)$-types, Q is a family of k-types, so the inductive hypothesis implies that this fiber is a $(k-n-2)$-type. Thus, all path spaces of $\mathsf{fib}_{(- \circ f)}(\ell)$ are $(k-n-2)$-types, so it is a $(k-n-1)$-type. □